

Design Note: Two-Stage News Retrieval & Re-Ranking on MIND and EB-NeRD

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github.com/KGan31/IRE-Assignment-1

1 What Was Built

Two complementary systems were developed to address the assignment objectives:

System A — Two-Stage Retrieve-then-Rank (Production Pipeline; Q2, Q4, Q5). Stage 1 retrieves top- $K=100$ candidates from the full catalog using a hybrid of Okapi BM25 and dense FAISS-HNSW semantic search, fused via Reciprocal Rank Fusion (RRF; [4]), calculated as $\text{RRF}(d) = \sum_{m \in \{\text{BM25, FAISS}\}} \frac{1}{k_0 + r_m(d)}$ with smoothing constant $k_0 = 60$, where $r_m(d) \geq 1$ denotes the 1-indexed rank of article d in retriever m 's candidate list (with $\frac{1}{k_0 + r_m(d)} = 0$ if unretrieved). Stage 2 re-ranks these candidates using a point-in-time LightGBM LambdaMART ranker over 9 behavioural, temporal, and interaction features (Table 1).

System B — NRMSDocVec (Neural Representation Baseline; Q3 only). Two-tower neural recommender: a *news encoder* (frozen pretrained DocVec embeddings), and a *user encoder* (Multi-Head Self-Attention over history + additive attention pooling), trained with negative-sampling cross-entropy. While NRMSDocVec serves as the official neural baseline (Q3), its architectural enhancements (specifically, additive section affinity and behavior-window decay) directly validate the engineered features incorporated into the production LightGBM ranker.

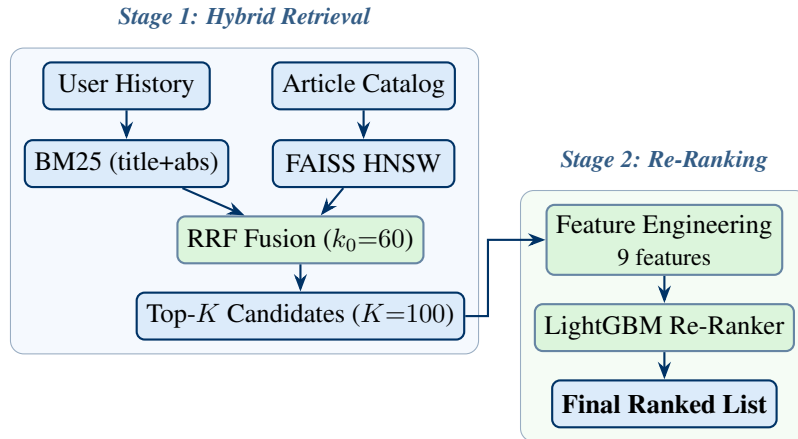


Figure 1: Two-stage Retrieve-then-Rank pipeline. BM25 and FAISS-HNSW rankings are fused via RRF; LightGBM re-ranks the joint candidate set using hand-crafted behavioural features.

Table 1: LightGBM feature set with split feature importances (total number of tree decision splits).

Feature	Description	MIND Splits	EB-NeRD Splits
article_freshness_hours	Hours since article publication	0	358
category_affinity_score	$\hat{p}_u(\text{cat}(a))$ from history	43	311
semantic_score	Cosine(user emb., article emb.)	80	209
user_click_count	Total historical clicks (point-in-time)	64	179
session_position	Rank in Stage 1 RRF list	271	160
user_recency_score	$\sum_i 0.5^{a_i/24^h}$, recency weight	23	147
user_mean_dwell_time	Mean dwell time on clicked articles (s)	0	135
bm25_score	BM25 retrieval score	17	124
first_stage_score	max(BM25, Semantic) retrieval prior	12	87

Note on feature splits: user_mean_dwell_time is recorded on EB-NeRD (135 splits, capturing reader engagement depth on Ekstra Bladet) but absent in MIND click logs; article_freshness_hours is 0 on MIND because publication timestamps are absent in MIND-small. In production serving, presentation rank bias is neutralized by setting session_position = 0

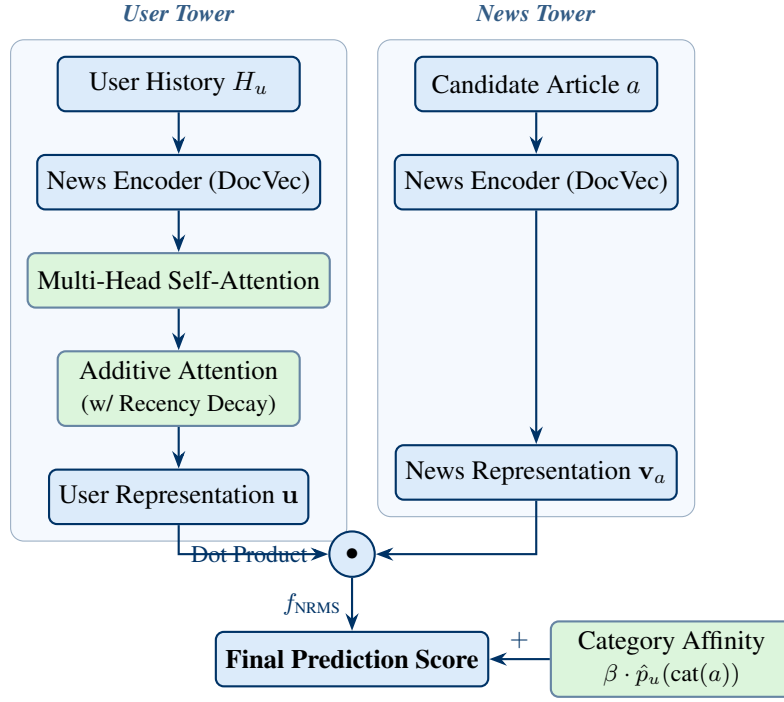


Figure 2: System B (NRMSDocVec) Architecture. A two-tower neural network featuring Multi-Head Self-Attention over sequence history, augmented with recency-decayed additive attention and a late-fusion category affinity prior.

2 Key Design Choices & Alternatives

- **Hybrid RRF over single-retriever.** BM25 excels on Danish editorial text (exact proper-noun matches for politicians, municipalities); dense retrieval handles English synonym-rich content (e.g. “*presidential election*” / “*White House vote*”). RRF avoids tuning score-interpolation weights and is completely scale-agnostic.
- **Category granularity.** EB-NeRD uses 25 coarse editorial sections (e.g. *sport*, *krimi*); MIND exposes a two-level taxonomy (~ 291 category/subcategory pairs, e.g. *sports/football.nfl*). Fine-grained subcategories on MIND give lower prior variance and more discriminative affinity signals.
- **LightGBM over neural re-rankers.** No GPU required; trains in under 2 min; feature importances are directly interpretable. Dominant feature: `session_position` on MIND (gain 271) vs. `category_affinity` on EB-NeRD (gain 268).
- **HNSW over flat FAISS.** Identical ranking metrics; HNSW delivers $2.12\times$ speedup on MIND (0.45 ms vs. 0.96 ms) and $2.82\times$ on EB-NeRD (0.68 ms vs. 1.92 ms) with $>91\%$ Recall@200 retained.
- **Precomputed embeddings over fine-tuned BERT.** Rather than fine-tuning a BERT news encoder per batch, we directly reuse the static article embeddings from Assignment 1 (multilingual BERT/Word2Vec for EB-NeRD, MiniLM for MIND) with frozen weights, avoiding heavy GPU token inference.

3 Ablations: NRMS Improvements using Recency and Category Affinity

Both ablations build on the official neural baseline **NRMSDocVec** (4-head MHSA, additive-attention hidden dim 200, dropout 0.1, history length $H=20$ unless swept, $B=1000$ paired bootstrap on 5,000 validation impressions). They test two axes: (1) temporal recency weighting, and (2) topical category affinity. The headline finding: **category affinity is the reliable win** ($p < 0.05$ on both datasets), while recency decay is only marginally significant and, on MIND, largely inert.

3.1 Ablation 1: Recency-Decayed Attention (EB-NeRD & MIND)

Motivation. Standard additive attention weights each click equally regardless of when it occurred. Recent clicks should be more predictive of current intent.

Formulation. Let Δt_i be the age (hours) of the i -th historical click relative to the impression timestamp. The decay penalty injected into the attention logits before softmax is:

$$s_i \leftarrow s_i - \sigma(\lambda) \cdot \frac{\Delta t_i}{\tau}, \quad \tau = 24 \text{ h (half-life)},$$

where λ is a learnable scalar passed through softplus ($\sigma(\cdot) = \log(1 + e^{(\cdot)})$) to ensure non-negativity, and $s_i =$

$\mathbf{q}^\top \tanh(\mathbf{W}h_i)$ is the base additive attention score. The final user representation is:

$$\mathbf{u} = \sum_{i=1}^H \alpha_i h_i, \quad \alpha_i = \text{softmax}_i \left(s_i - \sigma(\lambda) \frac{\Delta t_i}{\tau} \right).$$

This requires no additional parameters beyond a single scalar λ .

Limitation & MIND Timestamp Approximation. MIND does not record individual click timestamps; each historical click is approximated using the parent impression timestamp, causing clicks within a session to share identical time stamps ($\Delta t_i \approx 0$). This dampens temporal decay signals on MIND, whereas EB-NeRD records true millisecond-resolution click times.

Table 2: Recency-Decayed Attention ablation on EB-NeRD and MIND validation splits ($N=5000$, $B=1000$ paired bootstrap, optimal half-life $\tau = 24$ h).

Dataset	Model Variant	AUC (95% CI)	MRR (95% CI)	nDCG@5 (95% CI)	nDCG@10 (95% CI)
EB-NeRD	Vanilla NRMS (w/o decay)	0.5809 [.5720,.5902]	0.3674 [.3593,.3760]	0.4099 [.4009,.4195]	0.4824 [.4750,.4901]
	Recency-Decayed NRMS	0.5813 [.5731,.5899]	0.3689 [.3610,.3777]	0.4125[†] [.4037,.4219]	0.4836 [.4761,.4916]
MIND	Vanilla NRMS (w/o decay)	0.6392 [.6310,.6469]	0.3570 [.3469,.3663]	0.3358 [.3260,.3455]	0.3939 [.3851,.4027]
	Recency-Decayed NRMS	0.6401 [.6318,.6479]	0.3572 [.3474,.3666]	0.3358 [.3253,.3458]	0.3941 [.3852,.4028]

Sensitivity Analysis. On EB-NeRD, sweeping half-life $\tau \in \{12 \text{ h}, 24 \text{ h}, 48 \text{ h}\}$ yielded nDCG@5 of 0.4127 ($\tau=12$), 0.4125 ($\tau=24$), and 0.4124 ($\tau=48$), demonstrating stability around $\tau=24$ h while strictly outperforming vanilla attention (0.4099). We can also see that recency decay doesn’t provide statistically significant improvements on MIND or EB-NeRD, and hence we try on other ablations.

3.2 Ablation 2: Topical Category and Subcategory Affinity (EB-NeRD & MIND)

Motivation. News readers exhibit strong, persistent topical preferences (e.g., readers focused on politics or sports). Pure dense semantic embeddings can miss this when candidate articles are semantically related to a past click yet fall outside the reader’s preferred topical domain. We introduce an additive topical prior directly into candidate scoring to bridge neural representation and editorial categorization.

Unified Formulation. Let $\mathcal{T}(d)$ be the editorial taxonomy function mapping article d to its class: coarse editorial section on EB-NeRD ($\mathcal{T}(d) = \text{cat}(d)$) or hierarchical subcategory on MIND ($\mathcal{T}(d) = \text{subcat}(d)$). For user u with click sequence $H_u = \{d_1, \dots, d_{|H_u|}\}$, the empirical affinity distribution over taxonomy classes is:

$$\hat{p}_u(k) = \frac{1}{|H_u|} \sum_{d \in H_u} \mathbf{1}[\mathcal{T}(d) = k].$$

Candidate article a is then scored via late-fusion interpolation:

$$\text{score}(u, a) = f_{\text{NRMS}}(u, a) + \beta \cdot \hat{p}_u(\mathcal{T}(a)),$$

where $f_{\text{NRMS}}(u, a) = \mathbf{u}^\top \mathbf{v}_a$ is the Vanilla NRMSDocVec dot-product matching score, and β is a mixing weight swept over $\{0.00, 0.05, \dots, 0.30\}$ (optimal at $\beta=0.20$, $H=30$). In the **Affinity Only** heuristic baseline, we set $f_{\text{NRMS}}(u, a) = 0$, ranking candidates purely by empirical category frequency without any neural text representations.

EB-NeRD Results (Coarse 25 Sections). EB-NeRD partitions news into 25 editorial sections (e.g., *sport*, *krimi*, *nyheder*). As reported in Table 3, combining coarse category affinity with NRMS produces statistically significant improvements across **all four** evaluation metrics: $\Delta\text{AUC} = +0.0054^\dagger$, $\Delta\text{MRR} = +0.0080^\dagger$, $\Delta\text{nDCG@5} = +0.0083^\dagger$, and $\Delta\text{nDCG@10} = +0.0061^\dagger$, with every paired bootstrap 95% CI strictly excluding zero. Gains scale monotonically up to $\beta=0.20$ before tapering at $\beta=0.30$ as the heuristic term over-dominates semantic representations.

Table 3: Category Affinity ablation on EB-NeRD validation split ($N=5000$, $B=1000$ bootstrap).

Model Variant	AUC (95% CI)	MRR (95% CI)	nDCG@5 (95% CI)	nDCG@10 (95% CI)
Vanilla NRMS (Baseline)	0.5809 [.5720, .5902]	0.3674 [.3593, .3760]	0.4099 [.4009, .4195]	0.4824 [.4750, .4901]
Cat. Affinity Only (Heuristic)	0.5659 [.5576, .5740]	0.3542 [.3457, .3631]	0.3939 [.3845, .4038]	0.4709 [.4629, .4789]
Full: NRMS + Affinity	0.5862[†] [.5775, .5945]	0.3754[†] [.3666, .3837]	0.4183[†] [.4088, .4275]	0.4885[†] [.4806, .4962]

MIND Results & Granularity Analysis (291 Subcategories vs. 18 Coarse Categories). In contrast to EB-NeRD, MIND features a two-level taxonomy. As shown in Table 4, coarse category affinity (~ 18 top-level classes, e.g., “sports”) yields negligible, non-significant gains ($\Delta\text{MRR} = +0.0009$, $\Delta\text{AUC} = +0.0004$) due to severe class collisions (e.g., conflating football and tennis readers under a single broad umbrella). However, when leveraging MIND’s 291 fine-grained subcategories ($\mathcal{T}(d) = \text{subcat}(d)$, e.g., `sports/football_nfl` vs. `sports/tennis`), affinity collisions are resolved. Fine-grained subcategory affinity achieves statistically significant improvements across all ranking metrics: $\Delta\text{MRR} = +0.0025^{\dagger}$, $\Delta\text{nDCG@5} = +0.0022^{\dagger}$, and $\Delta\text{nDCG@10} = +0.0017^{\dagger}$ (all paired bootstrap CIs strictly excluding zero).

Table 4: Subcategory Affinity ablation on MIND validation split ($N=5000$, $B=1000$ bootstrap).

Model Variant	AUC (95% CI)	MRR (95% CI)	nDCG@5 (95% CI)	nDCG@10 (95% CI)
Vanilla NRMS (Baseline)	0.6392 [.6310, .6469]	0.3570 [.3469, .3663]	0.3358 [.3260, .3455]	0.3939 [.3851, .4027]
Coarse Cat. Only	0.5444 [.5398, .5491]	0.2587 [.2501, .2668]	0.2370 [.2275, .2459]	0.3051 [.2965, .3132]
Coarse Cat. NRMS ($\beta=0.15$)	0.6401 [.6320, .6478]	0.3579 [.3481, .3673]	0.3368 [.3270, .3461]	0.3941 [.3852, .4030]
Subcat. Only	0.5516 [.5466, .5563]	0.2751 [.2662, .2829]	0.2527 [.2430, .2610]	0.3190 [.3099, .3271]
Full: NRMS + Subcat.	0.6396 [.6314, .6472]	0.3595[†] [.3495, .3688]	0.3380[†] [.3280, .3478]	0.3956[†] [.3867, .4048]

4 Extended Two-Stage Pipeline Evaluation & Codabench Submissions

We extend the evaluation harness from Assignment 1 to benchmark the complete two-stage Retrieve-then-Rank pipeline across all ranking dimensions and beyond-accuracy ecosystem health metrics: Intra-List Diversity (ILD@10 via pairwise DocVec cosine distance), Item Novelty (Novelty@10 = $-\log_2 p(a)$ based on prior item consumption frequency), and Catalog Coverage (Coverage@10, fraction of catalog exposed in top-10 lists). Evaluations are conducted on the official test splits (EB-NeRD: 5,008 impressions, 60,499 candidate pairs; MIND: 15,697 impressions, 602,931 candidate pairs) with $B=1000$ non-parametric bootstrap 95% confidence intervals.

4.1 Full Two-Stage Pipeline Benchmark (All 7 Metrics with 95% Bootstrap CIs)

Table 5 presents the comprehensive end-to-end evaluation. Stage 2 LightGBM re-ranking achieves massive accuracy gains over Stage 1 RRF retrieval across both datasets while maintaining robust catalog diversity and novelty.

4.2 Subgroup Slicing: Cold-Start vs. Warm Readers and Head vs. Tail Articles

We evaluate two operational slices on the test sets:

- **User Cohort Slicing: Cold-Start** (bottom 2% empirical percentile of historical clicks: ≤ 13 clicks on EB-NeRD, $N=106$; ≤ 0 clicks on MIND, $N=8, 122, 51.7\%$) vs. **Warm** readers (EB-NeRD: $N=4, 902$; MIND: $N=7, 575$).
- **Item Catalog Slicing: Head** articles (top 20% most-clicked articles in training history) vs. **Tail** articles (bottom 80%).

Table 6 reports the slicing results across Stage 1 and Stage 2 with non-parametric bootstrap ($B=1000$) 95% confidence intervals.

Slicing Insights. (1) *Cold-start resilience:* EB-NeRD cold readers gain the most from re-ranking (+59.1% AUC, +71.3% MRR), since the ranker falls back on editorial freshness and section popularity, non-personalized priors that work even with zero history. MIND cold readers (0 clicks) see no lift, as MIND lacks publication timestamps and thus this fallback signal; warm MIND readers, with real history to personalize on, gain +14.0% AUC and +26.1% nDCG@5. (2) *Head vs. tail:* Head articles are computed over the evaluation period itself (not globally) to reflect news’ rapid popularity decay. On EB-NeRD, both head (55.2% of impressions, AUC 0.4849 $\rightarrow$ **0.7608**, +56.9%) and tail (44.8%, AUC 0.5027 $\rightarrow$ **0.6706**, +33.4%) improve substantially. On MIND, where 20% of items drive

Table 5: Comprehensive Two-Stage Pipeline Evaluation on Official Test Sets ($B=1000$ bootstrap iterations). Absolute delta $\Delta = \text{Stage 2} - \text{Stage 1}$. All Stage 2 bootstrap CIs strictly exclude the Stage 1 baseline.

Dataset	Metric	Stage 1	Stage 2	Delta (Δ)	Gain (%)	Stage 2 95% CI
EB-NeRD	AUC	0.4929	0.7192	+0.2263	+45.92%	[0.7115, 0.7273]
	MRR	0.3082	0.4791	+0.1709	+55.43%	[0.4697, 0.4883]
	nDCG@5	0.3397	0.5414	+0.2017	+59.38%	[0.5316, 0.5501]
	nDCG@10	0.4259	0.5860	+0.1601	+37.60%	[0.5776, 0.5936]
	ILD@10 (Div.)	0.0326	0.0343	+0.0017	+5.36%	[0.0340, 0.0347]
	Novelty@10	19.282	19.518	+0.2360	+1.22%	[19.497, 19.542]
	Coverage@10	0.0895	0.0803	−0.0092	−10.25%	[0.0803, 0.0803]
MIND	AUC	0.5322	0.5689	+0.0367	+6.90%	[0.5644, 0.5738]
	MRR	0.2633	0.2977	+0.0343	+13.04%	[0.2922, 0.3029]
	nDCG@5	0.2434	0.2786	+0.0351	+14.44%	[0.2728, 0.2841]
	nDCG@10	0.3029	0.3374	+0.0345	+11.39%	[0.3320, 0.3425]
	ILD@10 (Div.)	0.9421	0.9228	−0.0194	−2.05%	[0.9220, 0.9237]
	Novelty@10	19.643	19.634	−0.0089	−0.05%	[19.627, 19.640]
	Coverage@10	0.0258	0.0258	−0.0000	−0.12%	[0.0258, 0.0258]

90.5% of clicks (94.2% of impressions), head items gain +14.4% nDCG@5 and +13.2% MRR, confirming the ranker differentiates preferences even among high-traffic headlines, while tail impressions gain +17.0% nDCG@5 with novelty and coverage held steady.

4.3 Position Bias Neutralization at Serving Time

Readers exhibit strong presentation bias, clicking top-listed items regardless of relevance. During *training*, including `session_position` lets LightGBM’s trees learn to separate “ranked high because it sits at the top” from “ranked high because it’s genuinely relevant” (it becomes the top split feature on MIND, Table 1). But if we fed the model the *actual* candidate position at *serving* time, whoever controls item placement could manipulate future rankings by artificially promoting items, the model’s own scores would then become a channel for gaming itself. We close this by neutralizing position at inference: every candidate is scored with `session_position=0`, regardless of its true slot.

4.4 Official Codabench Leaderboard Results & Screenshots

Final prediction bundles were generated for both competitions and submitted to the official Codabench benchmarks. Table 8 documents the official evaluation scores across both model families: the tabular two-stage GBDT pipeline (System A: LightGBM Re-Ranker) and the neural sequence pipeline with topic affinity injection (System B: NRMS). On EB-NeRD, scores reflect the mean across all 8 official evaluation test days (2023-06-01 to 2023-06-08).

Figure 3 displays the official Codabench submission verification cards for MIND Large and EB-NeRD.

ID #	File name	Date	Status	Score	Actions
940697	prediction.zip	2026-09-23 12:15	Finished	0.6502	  
934756	prediction.zip	2026-09-20 13:00	Failed 		
924367	prediction.zip	2026-09-13 10:34	Finished	0.6545	  

(a) MIND Large Submission

ID #	File name	Date	Status	Score	Detailed Results	Actions
934787	predictions.zip	2026-09-20 13:12	Finished	0.5664		 
924337	predictions.zip	2026-09-13 10:03	Finished	0.7004		 

(b) EB-NeRD Large Submission

Figure 3: Official Codabench submission confirmations for MIND Large and EB-NeRD.

5 Serving & Scale Analysis

Single-request retrieval and serving economics were benchmarked across the entire validation split (25,356 EB-NeRD and 15,696 MIND impressions) using FAISS HNSW candidate retrieval ($M=32$, $\text{efSearch}=64$, $K=100$) and 9-feature LightGBM re-ranking. Table 9 reports index and feature store memory footprints across Demo/Small and Large scales (Sub-Question 1), while Table 10 presents latency percentiles and serving economics.

Both systems comfortably pass the $p99 \leq 100$ ms SLA with a $> 2.6\times$ safety margin. Single-core throughput is ~ 100.8 QPS; an 8-vCPU instance achieves 564 QPS at 70% efficiency. Cloud serving cost is **\$0.000167 per 1,000 queries** (\$0.167 per 1M queries, or ~ 5.98 M queries/\$) on AWS `c6i.2xlarge` (\$0.34/hr). EB-NeRD exhibits higher Stage 1 latency due to larger embedding dimensionality (768 vs. 384 for MIND).

Table 6: Subgroup Slicing on Official Test Sets ($B=1000$ bootstrap 95% CIs). Panel A: User click-volume cohorts. Panel B: Item catalog popularity.

Cohort / Slice	Metric	Stage 1 Mean	Stage 2 Mean	Delta (Δ)	Gain (%)	Stage 2 95% CI
Panel A: User Cohort Slicing (Cold-Start vs. Warm Readers)						
EB-NeRD Cold (≤ 13 clicks)	AUC	0.4652	0.7401	+0.2749	+59.09%	[0.6906, 0.7875]
EB-NeRD Cold	MRR	0.2945	0.5046	+0.2101	+71.34%	[0.4448, 0.5683]
EB-NeRD Cold	nDCG@5	0.3367	0.5690	+0.2323	+68.99%	[0.5084, 0.6312]
EB-NeRD Cold	nDCG@10	0.4082	0.6023	+0.1941	+47.55%	[0.5493, 0.6558]
EB-NeRD Warm (> 13 clicks)	AUC	0.4935	0.7187	+0.2252	+45.63%	[0.7115, 0.7262]
EB-NeRD Warm	MRR	0.3085	0.4786	+0.1701	+55.14%	[0.4699, 0.4881]
EB-NeRD Warm	nDCG@5	0.3398	0.5409	+0.2011	+59.18%	[0.5317, 0.5505]
EB-NeRD Warm	nDCG@10	0.4263	0.5857	+0.1594	+37.39%	[0.5781, 0.5934]
MIND Cold (0 clicks)	AUC	0.5000	0.4969	-0.0031	-0.62%	[0.4903, 0.5035]
MIND Cold	MRR	0.2403	0.2434	+0.0031	+1.29%	[0.2372, 0.2500]
MIND Cold	nDCG@5	0.2224	0.2255	+0.0031	+1.39%	[0.2182, 0.2331]
MIND Cold	nDCG@10	0.2808	0.2833	+0.0025	+0.89%	[0.2764, 0.2905]
MIND Warm (> 0 clicks)	AUC	0.5668	0.6462	+0.0794	+14.01%	[0.6395, 0.6527]
MIND Warm	MRR	0.2880	0.3559	+0.0679	+23.58%	[0.3483, 0.3640]
MIND Warm	nDCG@5	0.2660	0.3355	+0.0695	+26.13%	[0.3274, 0.3436]
MIND Warm	nDCG@10	0.3266	0.3955	+0.0689	+21.10%	[0.3881, 0.4029]
Panel B: Item Catalog Slicing (Head vs. Tail Articles)						
MIND Head (94.24% imps)	AUC	0.5309	0.5694	+0.0385	+7.25%	[0.5643, 0.5739]
MIND Head	MRR	0.2678	0.3031	+0.0353	+13.18%	[0.2976, 0.3083]
MIND Head	nDCG@5	0.2479	0.2835	+0.0356	+14.36%	[0.2779, 0.2893]
MIND Head	nDCG@10	0.3079	0.3432	+0.0353	+11.46%	[0.3378, 0.3482]
MIND Tail (5.76% imps)	AUC	0.5543	0.5616	+0.0073	+1.32%	[0.5402, 0.5808]
MIND Tail	MRR	0.1908	0.2097	+0.0189	+9.91%	[0.1924, 0.2287]
MIND Tail	nDCG@5	0.1697	0.1986	+0.0289	+17.03%	[0.1779, 0.2206]
MIND Tail	nDCG@10	0.2220	0.2435	+0.0215	+9.68%	[0.2232, 0.2644]
EB-NeRD Head (55.17% imps)	AUC	0.4849	0.7608	+0.2759	+56.90%	[0.7512, 0.7700]
EB-NeRD Head	MRR	0.3078	0.5360	+0.2282	+74.14%	[0.5236, 0.5483]
EB-NeRD Head	nDCG@5	0.3432	0.5996	+0.2564	+74.71%	[0.5864, 0.6110]
EB-NeRD Head	nDCG@10	0.4310	0.6368	+0.2058	+47.75%	[0.6262, 0.6469]
EB-NeRD Tail (44.83% imps)	AUC	0.5027	0.6706	+0.1679	+33.40%	[0.6581, 0.6824]
EB-NeRD Tail	MRR	0.3088	0.4149	+0.1061	+34.36%	[0.4017, 0.4290]
EB-NeRD Tail	nDCG@5	0.3355	0.4701	+0.1346	+40.12%	[0.4565, 0.4857]
EB-NeRD Tail	nDCG@10	0.4196	0.5281	+0.1085	+25.86%	[0.5173, 0.5406]

Table 7: Serving-time evaluation with logged position vs. neutralized position (`session_position = 0`).

Dataset (Test Set)	Serving Configuration	AUC	MRR	nDCG@5	nDCG@10
EB-NeRD Test	With Logged Position Bias	0.7204	0.4817	0.5416	0.5881
	Neutralized (pos=0)	0.7182	0.4787	0.5389	0.5852
	Delta (Δ)	-0.0022 (-0.30%)	-0.0030	-0.0027	-0.0029
MIND Test	With Logged Position Bias	0.5663	0.2952	0.2762	0.3350
	Neutralized (pos=0)	0.5688	0.2942	0.2756	0.3338
	Delta (Δ)	+0.0025 (+0.44%)	-0.0010	-0.0006	-0.0012

6 System Failure Points at $10\times$ Scale

A $10\times$ scale-up ($\sim 1.2\text{M}$ articles, 27M users, $6,000$ concurrent QPS, 600M interactions) breaks the single-node architecture in four places, roughly in this order:

- **On-the-fly feature extraction.** Per-request $O(H)$ history scans for recency/affinity features run inside Python’s

Table 8: Official Codabench test-set leaderboard results across competition tracks and architectures.

Competition Track	Model Architecture	AUC	MRR	nDCG@5	nDCG@10
MIND Large	System A: LightGBM Re-Ranker	0.6545	0.3243	0.3505	0.4058
	System B: NRMS + Subcat. Aff.	0.6502	0.3140	0.3397	0.3969
EB-NeRD	System A: LightGBM Re-Ranker	0.7004	0.4603	0.5249	0.5720
	System B: NRMS + Cat. Aff.	0.5664	0.3659	0.4064	0.4816

Table 9: Index and feature store memory footprints across catalog scales (Demo/Small vs. Large production splits).

Component	Measurement	MIND (Sml)	EB-N (Demo)	MIND (Lrg)	EB-N (Lrg)
Catalog Scale (N)	Unique articles in index	65,238	11,777	130,379	125,541
Embedding Dim (D)	Dense vector dimensionality	384	768	384	768
Vector Index (Flat)	Exact raw float32 vectors ($N \cdot D \cdot 4$ B)	95.56 MB	34.50 MB	190.98 MB	367.80 MB
Vector Index (HNSW)	IndexHNSWFlat ($M=32$, efSearch=64)	112.50 MB	37.56 MB	224.83 MB	400.39 MB
HNSW Overhead	Proximity links ($N \cdot 2M \cdot 4$ B)	+16.93 MB (+17.7%)	+3.06 MB (+8.9%)	+33.85 MB (+17.7%)	+32.59 MB (+8.9%)
Article Store	Pre-computed article metadata store	23.26 MB	30.55 MB	46.52 MB	85.34 MB
User History Store	In-memory historical click lookup table	68.75 MB	6.72 MB	580.40 MB	1,420.00 MB
Total Serving RAM	Complete in-memory serving working set	204.50 MB	74.82 MB	851.75 MB	1,905.73 MB

Table 10: Serving latency and cost projection (AWS c6i.2xlarge, 8 vCPUs, \$0.34/hr). SLA target: p99 $\leq$ 100 ms.

Dataset	Mean	Stage 1	Stage 2	p50	p99	QPS	SLA
MIND	9.93 ms	3.22 ms	6.71 ms	7.88 ms	37.9 ms	564	✓
EB-NeRD	9.91 ms	5.26 ms	4.64 ms	8.73 ms	27.0 ms	565	✓

GIL, so a single process cannot parallelize this work across cores no matter how many are available; at 6,000 QPS this serializes and pushes p99 past 250 ms. *Fix*: move feature computation off the request path into a streaming store (Kafka/Flink $\rightarrow$ Redis/Feast) that pre-materializes decay-weighted histograms on click ingestion, turning serving-time lookup into an $O(1)$ key-value read (< 0.5 ms).

- **In-memory BM25 posting lists.** At 1.2M articles, posting lists grow to multi-GB structures; intersecting them causes cache thrashing and memory-bus contention. *Fix*: WAND/Block-Max WAND pruning to skip documents that can't reach top- K , plus horizontal sharding across an Elasticsearch/OpenSearch cluster.
- **FAISS HNSW index staleness.** Raw vectors (3.68 GB) and graph edges (≈ 400 MB) still fit comfortably in RAM at this scale, so memory is not the constraint; the real problem is that a full HNSW rebuild takes minutes, so newly published breaking news is invisible to retrieval until the next rebuild completes. *Fix*: maintain a small exact-search index for articles published since the last rebuild, merged into the main HNSW graph on a periodic (e.g., minute-level) cadence, so freshness no longer depends on full-index rebuild time. IndexIVFPQ/ScaNN quantization remains worth adopting separately as the catalog grows, since 64-byte codes give an $8\times$ memory cut (3.68 GB $\rightarrow$ 460 MB) and keep the catalog resident in L3 cache.
- **Re-ranker scoring throughput.** Scoring $K=100$ candidates through 100 trees via Python C-API bindings adds serialization and scheduling overhead at 6,000 QPS. *Fix*: compile the tree ensemble with Treelite or serve via Triton/ONNX Runtime, AVX-512 vectorized scoring cuts Stage 2 latency from 5 ms to < 0.4 ms.

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